

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper B)

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- Socio economic profile of the region
- Unable to afford premium due to low income
- Unable to purchase because of unstable or volatile income sources
- People are not aware of the concept of insurance
- This may be on account of lower financial awareness in the region
- Low levels of financial literacy are often found in low income areas
- Residents may choose not to insure for some reason for example not having trust in insurance companies honouring the claims
- Alternatives to insurance such as support from family or government may be considered adequate
- Potential severity of claims may not be understood
- Lack of good insurance product
- The risks of not insuring may not be well understood
- People may not know how to buy insurance
- It may not be compulsory to buy insurance
- The companies providing the coverage in the region may not be financially strong companies to offer such products
- The premiums could be high due to high probability of claim since it is a flood prone area
- Policies come with high deductibles/excess , so decide not to purchase
- Home insurance may not be available as insurers are not interested in selling the product
- Cover may not be available for key risks that people want
- For example, flood may be considered uninsurable if flooding is a regular event

(½ Mark each)

[Max. 9 Marks]

Solution 2:

- a) Government – Direct intervention**
- Greater investment in flood risk management
 - Improve flood risk mapping
 - Provide direct financial support for reconstruction of dwellings in case of flood out of its revenue
 - Set up Flood disaster relief fund from out its tax revenues which will be managed for pay-out in case of floods
 - Levy a special tax to cover losses from such risks post facto an event.
 - Issue special flood bonds- a financial security where pay-outs of coupon or principal are based on frequency and severity of flood events
 - Government could run an advertising / public information campaign to encourage people to insure

- Allow new property development only if flood insurance is purchased

[4]

b) Through insurance market

- Encourage private insurers to offer Flood insurance products by providing tax breaks, lower capital requirement compared to other insurance products,
- Government could make such insurance compulsory across the country for all property owners with potential to cross subsidise high and low risk properties
- Tariff the product to protect consumers in high risk areas
- Price based on risk but subsidise customers directly in high risk areas
- Tax breaks for e.g., credit for premium in computation of income /other taxes,
- Set up a national reinsurance pool – a not for profit reinsurance pool to keep re-insurance available and affordable at high risk areas
- Make it mandatory to offer flood coverage with property insurance
- Minimum premium targets for each insurer based on their market share
- Allow insurance companies to issue CAT bonds (some form of alternative risk transfer)- financial instruments where pay-outs are linked to losses from floods i.e., pay-out will be based on whether the event occurs and the extent of losses)

(½ Mark each)

[4]

[Max. 8 Marks]

Solution 3:

a)

Product features as per the need of the customers

- Offer ranges of sum insured to cater to even small home dwellers
- Set level of excess so as not to entertain small claims that would be uneconomical to administer
- the product should provide a reasonable level of coverage so should have a reasonable deductible,
- otherwise there would be no benefit of buying insurance
- Could offer a long-term product so as customers are freed from the hassle of yearly renewal (for e.g., 10year product covering structural risks or longer)
- Allow for some level of cross subsidy between policies sold in high risk areas and low risk areas
- cross-subsidies between different regions may not be sustainable in the long term as the risk of anti-selection will definitely arise
- Could accept a lower profit margin for the product overall
- However for a private insurer to be sustainable it needs to meet the risk adjusted return on the capital deployed to support the product
- Any regulatory constraints on the differential pricing needs to be considered

[1/2, max 5]

b)

Product Marketing

- There might be low awareness of the insurance in the region
- Can take steps to educate about the needs for insurance
- This could also be done through any industry body
- At the same time marketing the product through various cost effective measures
- For e.g., Including using satisfied claimants in the advertisements
- Simple product design , clarity in policy wordings

The sales of the product

- For example, through door-to-door agents
- For example, through another business operating in the community (such as the Post Office)
- If the internet is accessible it can also sell through its website
- Through direct sales effort like cold calling the customers
- Take out ads in local language with locally popular people

(½ Mark each)

[Max. 4]**[9 Marks]****Solution 4:**

- Flood zone the property is located in
- Type of coverage (building and/or contents insurance)
- Size of the dwelling
- The deductible chosen, if any
- Location of contents (for eg., elevated or not)
- Property type of construction (replacement cost)
- Property age
- Number of floors
- Primary/Non-Primary residence
- Loss history

(½ Mark each)

[Max. 5 Marks]**Solution 5:**

- Shows the financial effect of differences between actual experience and that assumed in the valuation.
- Shows the financial effect of writing new business
- Deviation or inadequacy in the pricing basis
- Provides a check on the valuation process if performed independently
- It could be a source of reconciliation between opening and closing balance sheet
- Provides information required for remuneration of sales or management
- Provides information to aid understanding of sources of surplus
- example identifying if there are unprofitable contracts so that remediating actions can be put in place
- It might be requirement to produce analysis of surplus on annual basis by regulator
- It might be required by board to perform analysis of surplus at least on an annual basis

(½ Mark each)

[Max. 4 Marks]

Solution 6: Surrender

- Asset share: Surrender value should give regard to the accumulated value of the policy at early durations
- At later duration it should be consistent with the maturity values
- The amount of profit one would like to take from the surrendering policy
- The policyholder would expect that a reasonable surrender value scales are provided and therefore should meet policyholder reasonable expectation
- Ease of calculation: The surrender value should be easy to calculate and explain
- Consistency with surrender values and the methodology used in the past for similar plans
- System constraints for implementation of the surrender value methodology as complex methodology may not be supported by core policy system
- or requires significant system development
- To compare the surrender value provided with the competition to ensure that it does not lead to reduced marketability of the product
- or reputational risk for the insurer
- Regulatory constraints or regulation which may provide floor on the level of the surrender values

(½ Mark each)

[Max. 5 Marks]

Solution 7:

- When was the surrender scale last reviewed?
- Analysis of experience- surrender by policy duration of surrender, age, ppt, premium band, policy term, to understand where actual losses are coming from
- Is loss arising in certain model points or across the board
- For e.g., if it is at early durations is SSV scale set at very high levels compared to AS
- Or if it at a later duration- it is being set at a level inconsistent to maturity value
- Is SSV scale calculated for an average model point which for e.g., might be proving to be overgenerous for low premium policies or at higher ages
- What is the methodology used in calculation of SSV (for e.g., AS, BE reserve at that point of time or prudent reserve at that point of time or some other method) or
- Is a blended method used, if yes,
- how does it relate to AS at earlier durations and maturity value at later durations?
- Is the company making less profits on surrender as compared to a comparable in force policy?
- Review of current investment strategy, Investment assets backing the product- fixed interest securities, volatile assets?
- Did the company have adequate liquid assets/premium inflow to manage surrender pay outs or was it liquidating assets to meet pay-outs
- How does SV compare to Market value of assets at the time of surrender?
- Is there a dynamic MVA allowed for at the time of surrenders?
- How does it compare to competition?
- Dynamic policyholder behaviour- is there any correlation between surrender rate and external economic environment like unemployment rate, inflation rate, interest rate
- Surrender experience by channel to understand if the issues are more channel specific- for eg., a particular distributor seeing large surrenders than expected
- Other analysis-
 - i. Existing system constraints which prohibit use of a better methodology/ approach
 - ii. Any operational issues- leading to an error in calculation
- regulatory constraints which are causing the loss

(½ Mark each)

[Max. 10 Marks]**[50 Marks]****Solution 8:**

Reasons of offering

- Increase business volumes
- Venture in new age products since insurer is heavy on traditional products, a step towards transformation
- Marketing: since lot of people take daily cabs and offering unique insurance cover will increase its brand awareness
- Cross sell other products by using database of customers
- Profiling on customer based on their riding habits/usage and use this as a socio economic factor in pitching other products
- Diversification across distribution/products
- Compete with new age insurers
- Compensate for price competition observed on traditional portfolio
- Experiment with 1 provider before going big
- Lack of other opportunities to deploy capital or existing capacity
- Bundled product with every ride will reduce risk as compared to offering it on stand alone individual basis (reduced anti-selection)
- No need of underwriting, straight through process as it a simple product accident product
- Lower expenses in managing as product is sold on group basis
- First mover advantage

(½ Mark each)

[Max. 5 Marks]**Solution 9:****a) Benefit Design****(3)**

Accident death/disability benefit

- Accident benefit is simple benefit but risks relate to inability to prove accident as cause of claim.
- What if claim was due to carelessness of rider –for eg., not wearing a seat belt
- Definition of disability needs to be made extremely clear.
- For eg., Disability needs to be defined appropriately in terms of inability to carry out any occupation etc if the intent is to keep cost to a minimum or
- A loose definition would end in payouts not allowed for in pricing

Hospitalization reimbursement-

(2)

- again difficult to prove cause was due to accident (for eg., not a pre-existing condition – neglected fracture, head injury etc.)
- Small and petty claims (for eg., sprain) leading to difficulty in administration

Trip Delay

(2)

- Trip delay is a risky cover. Risk of collusion between rider and cab driver.
- A driver might indulge in fraudulent behaviour and takes more time to complete ride

- The coverage could have high seasonality and will be affected by various factors.. e.g. high losses during rainfall, foggy season, peak hours or in certain congested areas
- 1 hour time maybe very low threshold for claim to be triggered
- Time taken after booking and ride onboarding is also covered under 1 hour duration, since the delay is counted from time of booking, hence
- high Moral hazard risk
- small claim sized which are unviable o administer

Damage to lap top

(1)

- Proving that the damage was due to an accident
- Fraudulent behavior by customer in overstating the value of lap top
- claiming replacement even when the device can be repaired

(½ Mark each, Max. 8 Marks)

b) Other risks for XYZ GI

- Lack of data and experience to price, design and administer such a product
- Too much dependence on single aggregator/customer
- Company offers traditional product, so system integration of such products might be challenging and expensive for a single provider.
- Risk of fraud by cab aggregator/driver or riders
- Loose policy wordings might cause increase in claims which are not accounted for
- Legal risk if contract with aggregator or customer is poorly worded
- Competition may soon launch a similar product and undercut , resulting in loss of business
- Pricing/reserving risk- Inadequate price /reserves
- Rejection of claims may lead to loss of reputation and impact other business
- Profitability / Capital strain due to higher than anticipated losses
- Risk outweigh the small uptick in business for a large company
- Thousands of coverages/and probable high number of claims a day would require a strong IT system and hence upfront investments
- Difficult to arrange reinsurance for such cover or any other risk transfer arrangements
- Accumulation of losses due to traffic congestion in a particular area
- There is also a risk of intervention if the new product becomes subject to any additional regulatory attention.
- Does not lead to level of cross sell as anticipated
- Increased litigations even for small claim amounts
- Higher claim management costs for eg,, from small size of claims in case of trip delay

(½ Mark each, Max. 8 Marks)

[16 Marks]

Solution 10:

Data Requirements

- Data is essential to estimate frequency in case of all coverages and severity amount of claim in case of hospitalization

Internal data

- Company is large General insurer and lot of data must be available around
 - Frequency of personal Accident
 - % if multiple claims from a single accident (multiple riders in a cab ride)

- Frequency of hospitalization due to accident
- Severity of cases arising from accident injury claims in case of hospitalizations
- Ideally the information can be analysed by geography to allow for differences in experience
- Past data of same can be used and adjusted for
 - Trends (inflation, demographic changes)
 - Geography mix of cab operations
- Laptop replacement costs
 - Market information on lap top prices
 - Most popular brands of laptops
 - Past inflationary trends in prices of laptops

Hence the severity so derived need to be adjusted for inflation in replacement costs.

External data

- Competitor/industry data in respect of personal accident/hospitalization frequency and severity from accidents.
- This needs to be adjusted to allowed for inflationary impact, differences in underwriting or claims policies.
- Lot of data can be sourced from cab aggregators
 - Geography of operations
 - Size of exposures and how it could vary in future
 - Past accident records
 - Police complaints
 - Driver details rating factors
 - Age
 - Years of experience
 - Rating of driver
 - Past incidents
 - Education
 - Own vehicle or not. Etc.

The data so sourced needs to be adjusted for known and expected changes

- in geography of operations,
- in vehicle profile (for eg., age of cabs allowed)
- driver profile
- driving practices for eg., policy relating to remote monitoring of driver speed
- Any changes in law relating to traffic violation fines, traffic enforcement which could change risk profile
- - Technology used to set time estimate for trip and how delay is measured
 - Past data if any on time taken by a cab to complete rides across various drop points across cities which will be covered,
 - % of cases where it exceeds estimated time
 - Average size of bills typically split by geography
 - The data needs to be adjusted for expected impact of increase in cab fares due to inflation /government ordered change
- External data can be sourced
 - Any published report on cab accidents

Which needs to be adjusted to allow for risk management practices of cab aggregator

- Data from other countries with similar product
- Both need to be suitably adjusted to suit the risk profile of local country

(½ Mark each)

[Max. 12 Marks]**Solution 11:****Factors influencing final premium charged per ride**

- Final Premium could be different from calculated premium as it will be loaded for expenses, cost of capital, profits, contingency margin[1]
- Other factor influencing final premiums are
 - First mover advantage will enable company to charge higher premiums
 - Backing of strong free asset position will influence final premium
 - Relationship with cab aggregators and opportunities to cross sell will have a major influence
 - Cover is available on all rides, it might lead to huge business opportunity and company might be willing to load lower margins
 - Whether competition is pitching for similar product to enter in market
 - Whether the product is positioned internally as a loss leader to gain other business
 - A lower price might attract other cab aggregators
 - Regulatory environment/restrictions
 - Premium paying capacity of cab aggregator
 - Taxation

(½ Mark each)

[Max. 6 Marks]**Solution 12:****Pros**

- It is an acknowledgement of a manager's performance and their contribution to the company
- By deferring the vesting date it enables the company to retain its talent
- Encourage employees to acquire ownership in the form of shares
- Align interests of employees and shareholders
- Employees who are also shareholders will be invested in company performance and growth
- If new age companies are offering ESO it might be a good HR policy to offer the same
- Defer immediate cash outgo – now for a large company cash outgo might not be an issue, but
- if the company is looking at some new opportunities/business strategy which requires money, ESO might be a best way to compete with new age players as cash out go is deferred, and it can be utilized in more profitable avenues [1]
- The 3 year period after vesting gives some flexibility to employees to decide on when to exercise the option.

Cons

- Stock options can be difficult to value
- The vesting period is 5 years. There might be no market reference point as there may be no comparable exchange traded options which may be traded for that period of time. [1]
- Setting the strike price and terms of option requires expert input from investment experts with expertise in option pricing etc.
- It can be an expensive scheme to run
- Lack of appreciation of the compensation by the employees as it just too complicated and

- deferred benefit, employees may appreciate immediate cash in their hands more
- Expenses and effort need to be incurred to teach employees to appreciate the compensation
- If ESO is not priced properly, it can lead to a situation where managers prosper even if the company is doing poorly
- Complicated tax issues for employee
- If the share price of the company's shares does not increase and the employee feels they have no control over the share price outcome, then it can affect morale and retention;[1]

[1/2mark each unless stated otherwise]

[Max 11 Marks]

[50 Marks]
